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POSITIONING, NAVIGATION AND TIMING (PNT) LABORATORY

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DEPARTMENT OF
SCIENCE & TECHNOLOGY

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Detection & Classification Of Spoofing Attacks On GNSS Signals Using AI-Based Models On NavIC.

Name: Yunes Abdulghani Mohammed Ghaleb

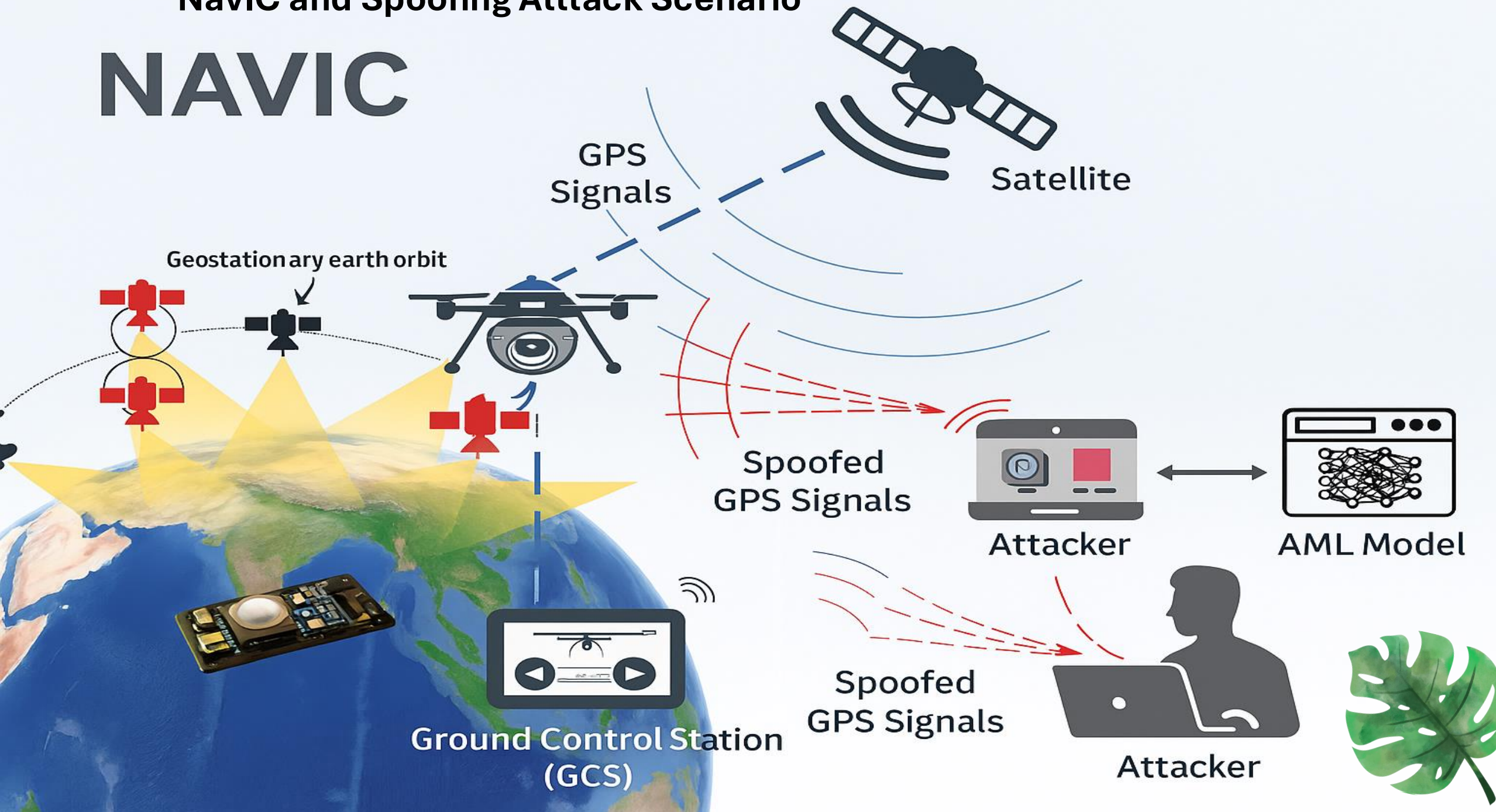
USN: 23BCAR0626

Specialization: BCA-AI



NavIC and Spoofing Attack Scenario

NAVIC





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Introduction

GNSS (Global Navigation Satellite Systems) provides positioning, navigation, and timing services.

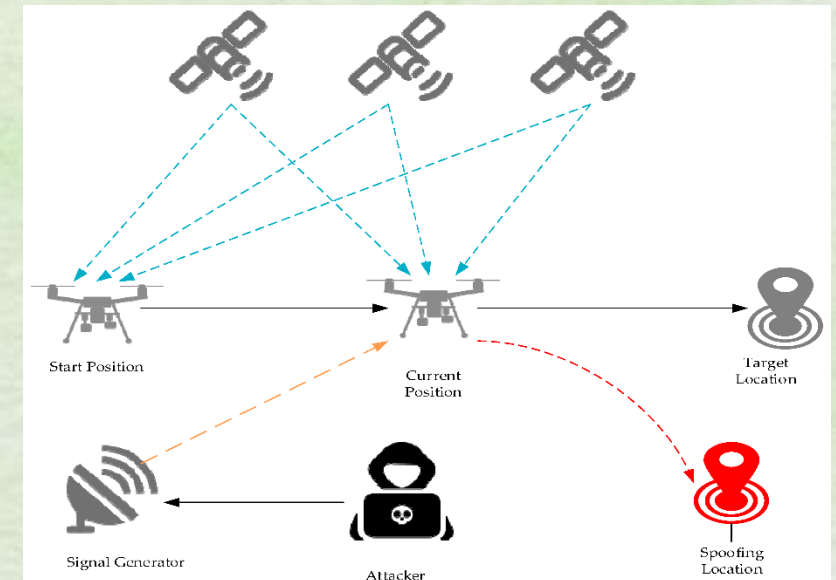
PNT: Positioning, Navigation, and Timing are critical for modern life, including transport, communication, and security.

The NavIC system is very important for navigation and security in our region. However, it is vulnerable to "spoofing attacks".

Spoofing attacks: sending fake GNSS signals to deceive the receiver.

This can cause incorrect positioning, leading to serious security risks.

AI can play a key role in detecting and preventing spoofing attacks.



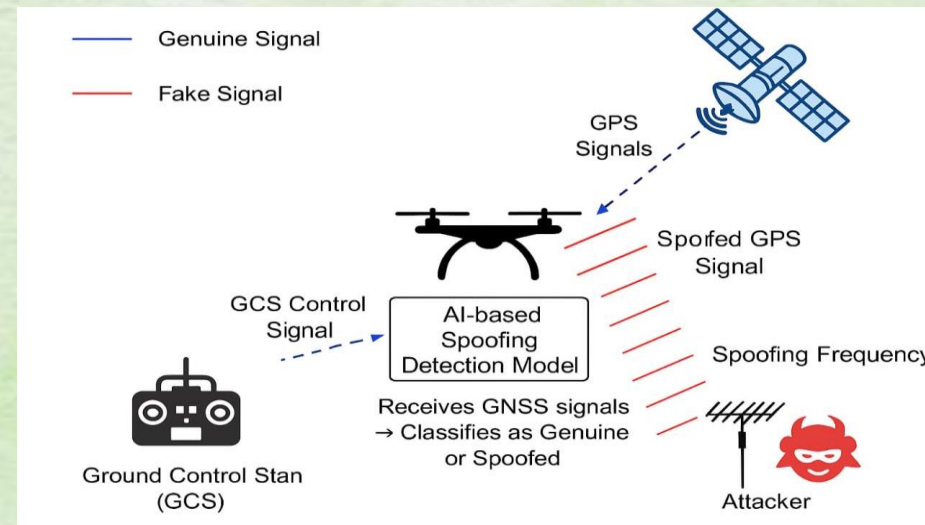


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Project Objectives

**Develop and compare three AI models (DNN, Random Forest, and XGBoost).
Use real-world GNSS data to classify authentic vs. spoofed signals.
Identify the most reliable model for practical deployment.**





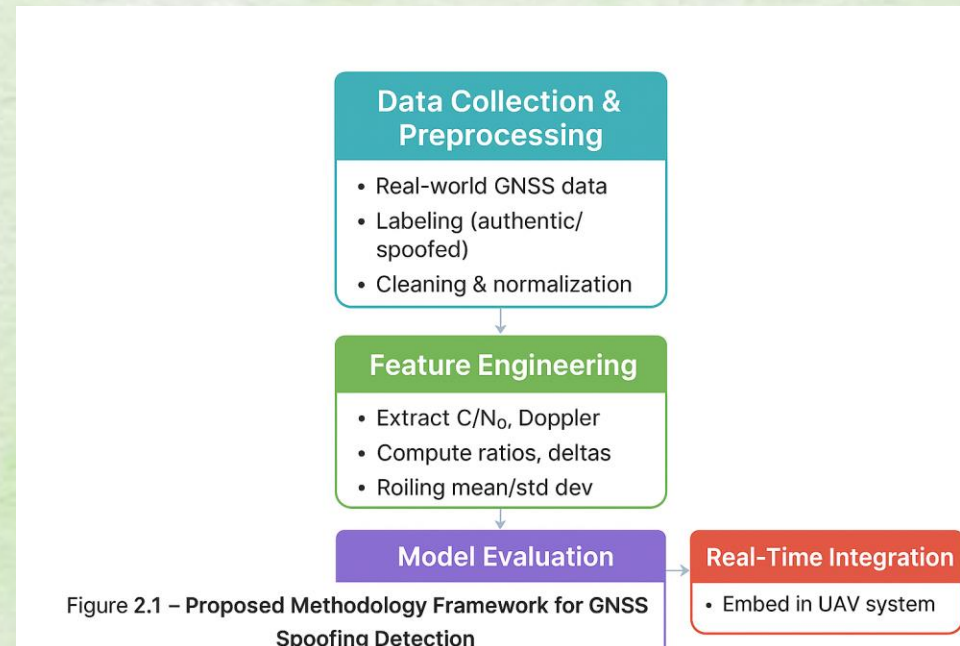
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Proposed Methodology

Data Collection → Feature Engineering → Model Training → Evaluation

XGBoost outperformed DNN and RF with 95.5% accuracy and lowest False Negatives.



Model Development

Data Preparation:

Real GNSS signal data collected, labeled, and normalized for consistency.

Feature Engineering:

Extracted key metrics (C/N_0 , *Doppler*) and created derived features (*CNO_Ratio*, *Rolling Mean*) to enhance detection.

Model Training:

Three models — *DNN*, *Random Forest*, and *XGBoost* — trained and compared under identical conditions.

Evaluation:

Models assessed using *Accuracy*, *Recall*, *Precision*, and *F1-Score* on real test data.

Key Result:

XGBoost achieved 95.5% Accuracy and minimized False Negatives to 334, proving the most reliable for GNSS spoofing detection.





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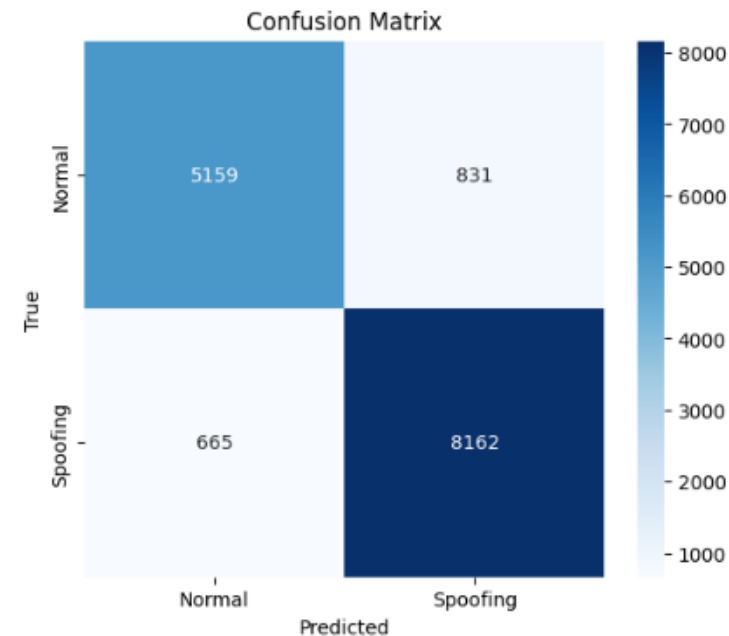
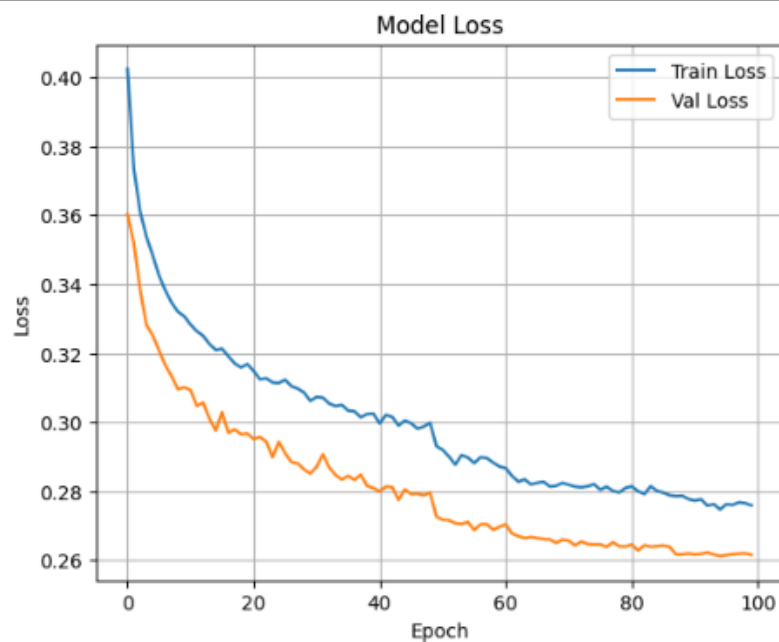
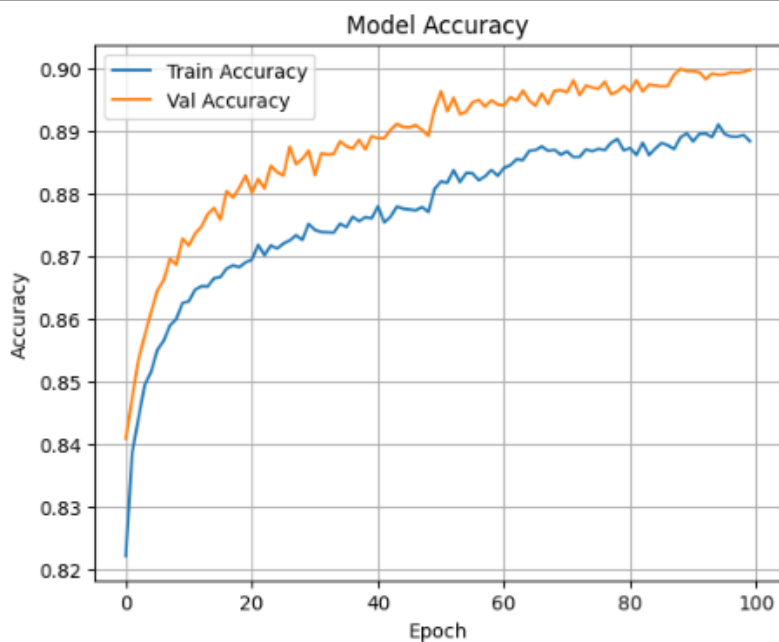


Results

Model	Accuracy	Precision	Recall	F1-score
DNN / ANN	0.90	0.90–0.92	0.90–0.92	0.90–0.92
Random Forest	0.93	0.93	0.93	0.93
XGBoost	0.95	0.95–0.96	0.95–0.96	0.95–0.96

XGBoost achieved the highest accuracy and lowest false negatives, proving most reliable.

DNN Model



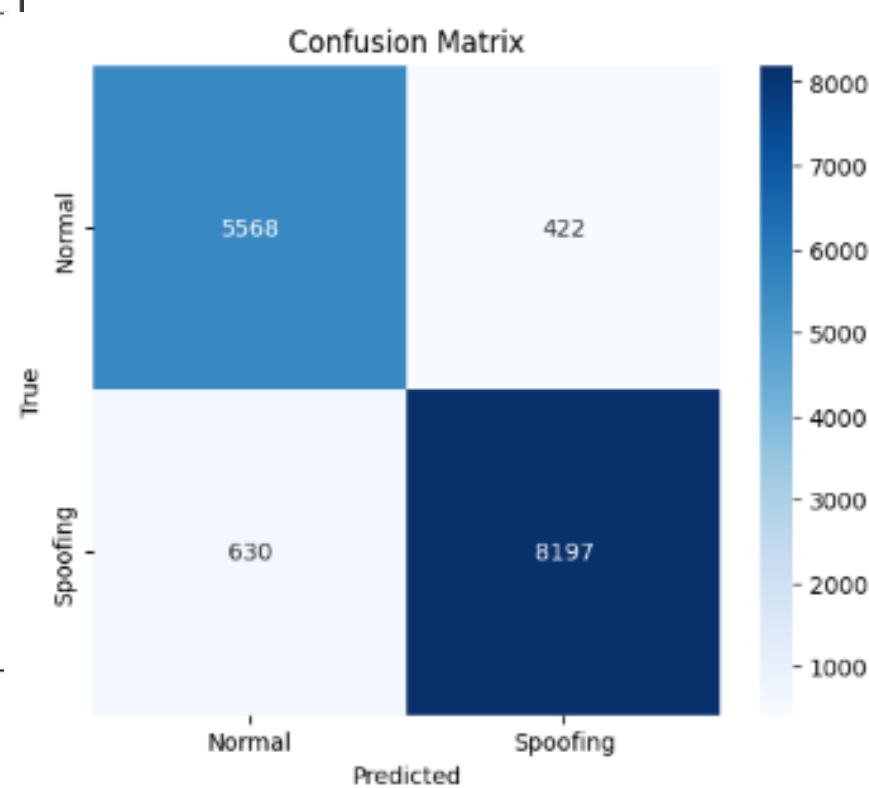
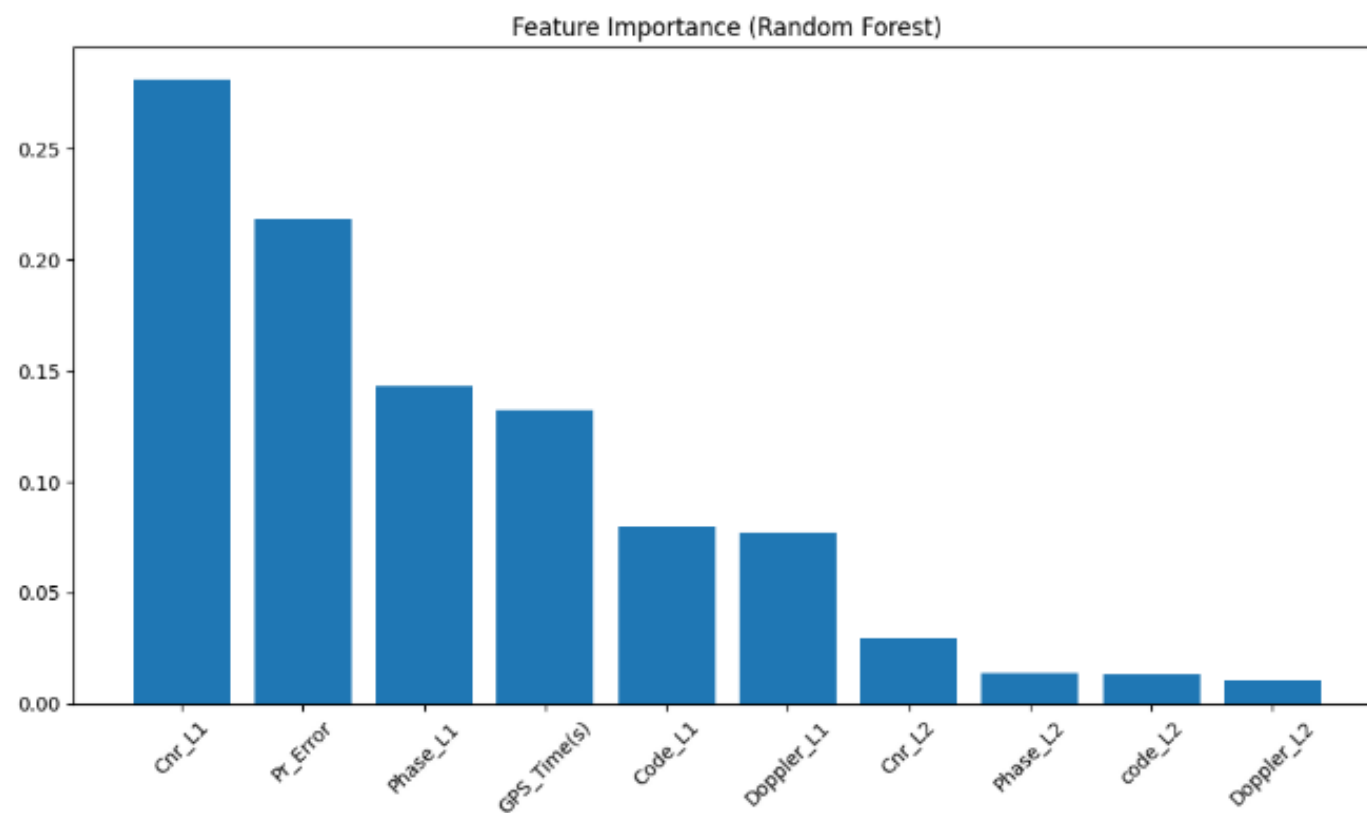


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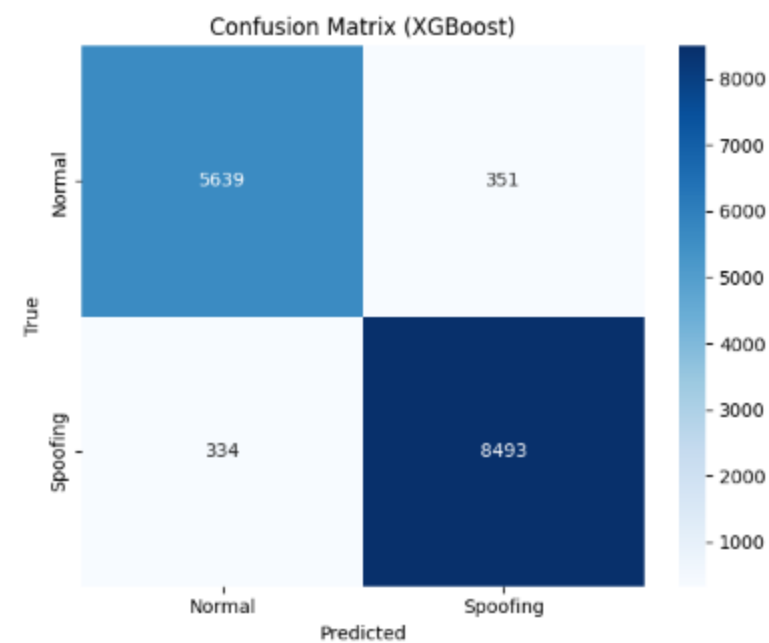
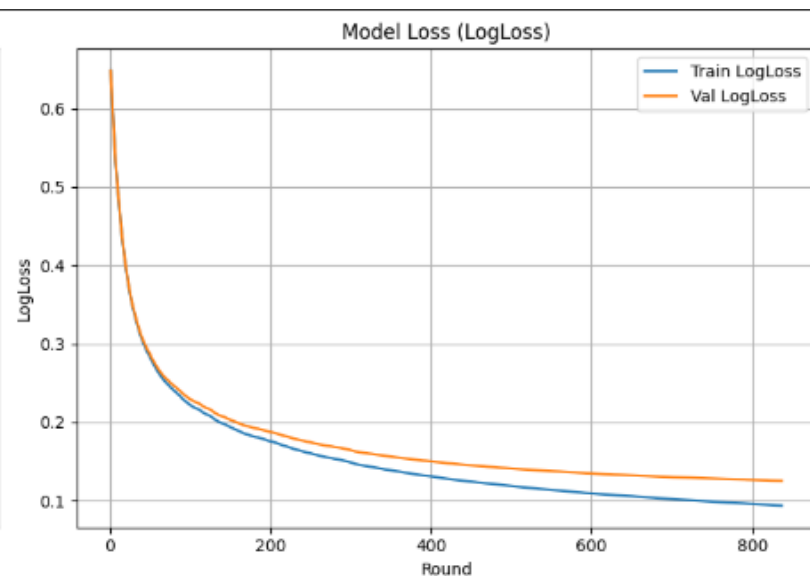
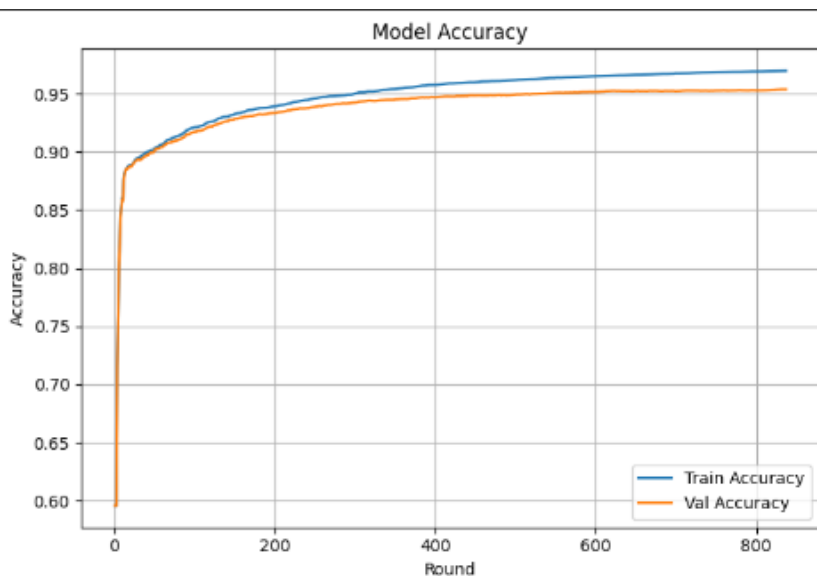


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Random Forest Model



XGBoost Model



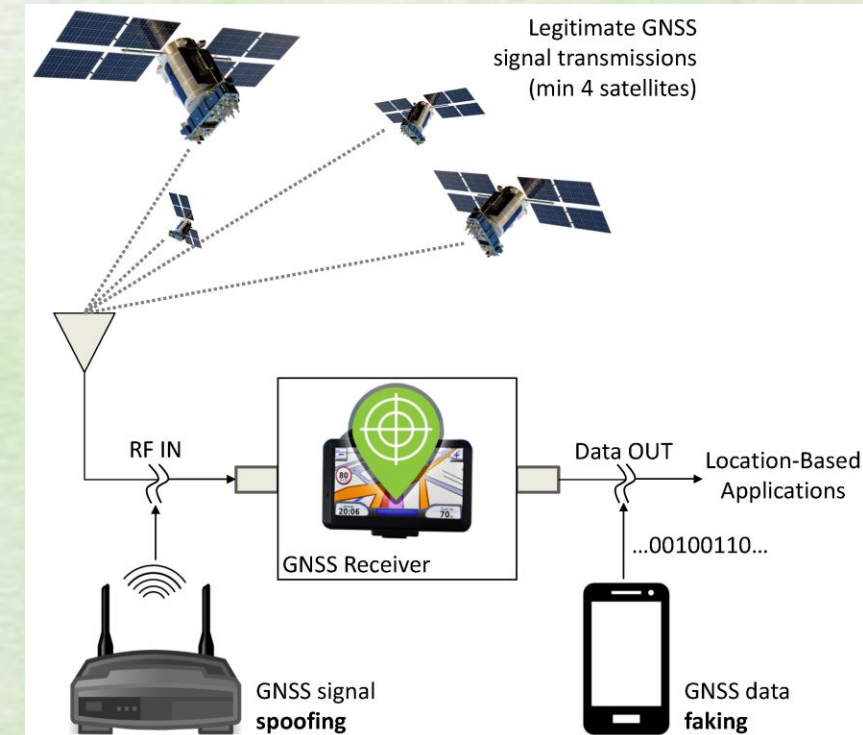


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Discussion

**XGBoost showed superior accuracy and recall under real GNSS data.
Minimizing False Negatives (FN = 334) was key for security.
DNN generalized well; RF was interpretable but less sensitive.
XGBoost is ready for real-time UAV deployment.**





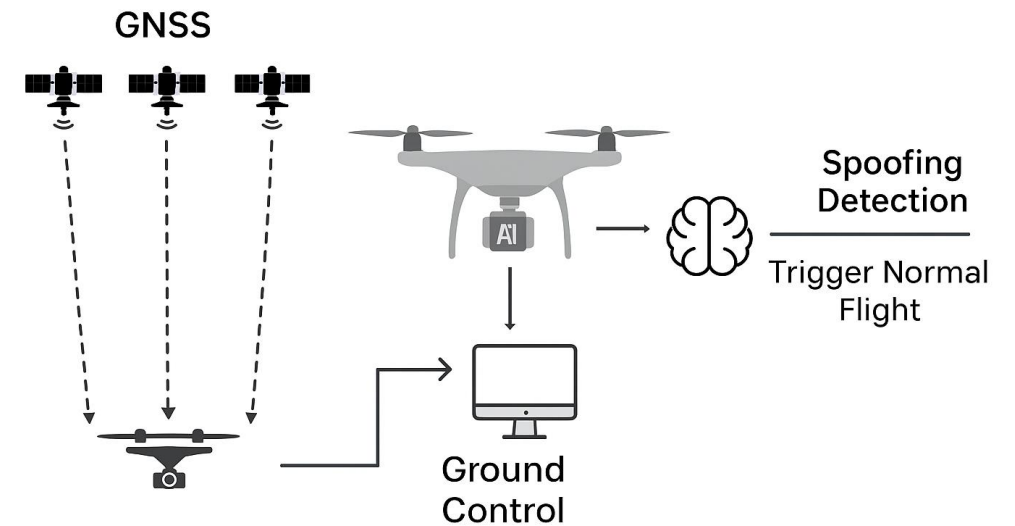
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Real-Time Implementation Plan

Embed XGBoost in UAV onboard processor.
Analyze GNSS signals in real-time.
Trigger alerts or switch to INS when spoofing detected.
Validate performance through controlled field testing.

Real-Time Spoofing Detection





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Future Work

Integration of trained AI models into drone navigation systems.
Field testing with real GNSS signals under controlled spoofing/jamming attacks.
Improving models for real-time detection and decision-making.
Exploring ensemble and hybrid approaches for enhanced robustness.





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Conclusion

This study successfully developed and evaluated three AI models—DNN, Random Forest, and XGBoost—for reliable detection of spoofed GNSS signals using real-world data. The results underscore the critical importance of diverse and representative datasets in building trustworthy AI-driven security systems. Moving forward, the next phase will focus on real-world integration and field testing on dynamic platforms, validating the models under live spoofing and jamming scenarios to ensure robust and operationally resilient performance.





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